1. Write a Python program using nested loops to print the following pattern:

   1

   12

   123

   1234

   12345

2. Write a Python program using nested loops to find and print all prime numbers between 1 and 100.

3. Given two lists, use nested loops to find and print the common elements:

   list1 = [10, 20, 30, 40, 50]

   list2 = [20, 40, 60, 80, 100]

4. Write a Python program to count the frequency of each character in a string without using count().

5. Write a Python program to find the first non-repeated character in a string.

6. Write a Python program to find the second largest unique number from a list without using sort().

7. Write a Python program to remove duplicate elements from a list while maintaining the original order.

8. Given a dictionary of students and marks, find and print the student(s) who scored the highest marks.

9. Write a Python program to count the frequency of each word in a sentence and store the result in a dictionary.

10. Given two dictionaries containing product names and prices, merge them into one dictionary. If a product exists in both, keep the higher price.

11. Create a function calculator(a, b, operation) that performs addition, subtraction, multiplication, and division based on the given operation.

12. Create a function that accepts a list of numbers and returns the largest number, smallest number, sum, and average.

13. Create a BankAccount class with account holder, account number, and balance. Add methods for deposit, withdrawal, and displaying balance. Prevent withdrawal when the balance is insufficient.
14. Create an Employee class with employee name, ID, department, and salary. Add methods to display details, increase salary by a given percentage, and check whether the salary is above ₹50,000.
15. Create a Vehicle class with brand and speed. Create Car and Bike child classes with additional attributes. Use inheritance and method overriding to display their complete details.